

PRANAV GANESH CHANDRATRE

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PROFILE

MSc Data Science candidate doing applied research on diffusion models for medical imaging. Sole contributor on an MSc thesis that LoRA fine-tunes a latent diffusion backbone in PyTorch and measures DDIM-inversion uncertainty and sampler efficiency against MC-Dropout baselines. Took a multimodal audio-to-structured-data system from hackathon prototype to a merged pipeline submitted for clinical pilot, and built tested data pipelines for model training and evaluation. Earlier physics background: An MSc in Physics with thesis and two-year research appointment training graph, dense and residual neural networks on roughly 100,000 Geant4-simulated CERN Compact Muon Solenoid HGCAL detector events across multi-GPU Tesla P100 hardware. Interests: generative media, inference optimization, statistics of model evaluation.

Work authorisation: Indian national on a UK Student visa; eligible for the UK Graduate route on completion of the MSc. Open to London or Zürich, with sponsorship for either.

EDUCATION

University of Birmingham — MSc Data Science | Sep 2025 – Sep 2026 expected

Thesis: Uncertainty Quantification in Diffusion Models for Chest Radiography.

IISER Kolkata — BS-MS Dual Degree, Physical Sciences | 2017 – 2022

Specialisation in Data Analysis and Machine Learning; GPA 8.52/10. INSPIRE Scholarship, Department of Science and Technology, Government of India (2017–2022).

SKILLS

ML Frameworks: PyTorch, Hugging Face diffusers, PyTorch Geometric, scikit-learn, LightGBM, XGBoost, statsmodels, TensorFlow (coursework and MS thesis)

Generative & Multimodal: latent diffusion (LoRA fine-tuning, DDIM inversion, sampler tuning), Generative Adversarial Networks, Convolution Neural Networks and U-Net, Graph Neural Networks, Residual and attention mechanisms, speech-to-text and LLM structured extraction, Agentic AI, Reinforcement Learning, Transformers and Sequence Models.

Data Pipelines & MLOps: Python, SQL, dbt, BigQuery, Docker, Git, GitHub Actions CI, Hugging Face Hub

Statistics & Evaluation: negative controls, Friedman and Wilcoxon tests with BH correction, rank-biserial effect sizes, Negative Binomial GLMs, , rolling-origin cross-validation, FIDs, AUPCR, MC Dropout, F1-score, k_cross validation

Languages: Python, SQL, C, C++, MATLAB

RESEARCH & WORK EXPERIENCE

MSc Thesis — Uncertainty Quantification in Diffusion Models for Chest Radiography | June 2026 – Sep 2026

University of Birmingham | Sole contributor: architecture, experiments and thesis | Code+Report: <https://github.com/pranavgc/Uncertainty-Quantification-in-Diffusion-Models-for-Chest-Radiography>

Stack: PyTorch, Hugging Face diffusers, latent diffusion (RoentGen-class), DenseNet-121, NumPy/SciPy/statsmodels | Data: VinDr-CXR, 5 of 22 annotated findings, 512×512 downsampled from 2000×2000 | Compute: 2 × T4, 16 GB each

- LoRA fine-tuned the latent diffusion backbone (rank 8, label-free captions, 2,000 in-domain radiographs) as a pre-registered intervention, raising edge-dominance by $\Delta p = +0.046$ to $+0.061$ ($p < 0.0001$) and inverting the domain-gap hypothesis; full fine-tuning in progress.
- Built an inversion-anchored uncertainty estimator that DDIM-inverts real radiographs to a fixed latent ($\eta = 0$) and takes per-pixel variance over 20 stochastic reverse passes, so each estimate describes one input image rather than the generator's overall diversity.
- Cut denoising steps from 30 to 15, halving the sampler passes behind a 39.6 s per image generation on 2 × T4, which cut variance 32.4% for +0.016 localisation AUROC; halving η cut variance 47% for +0.037. An N = 10 negative control, varying estimator sample count alone, returned null on both, placing the effect in the sampling process.
- Showed most raw variance carries no localisation signal, tracing it to high-contrast anatomy: the highest-variance pixel fell inside an annotated lesion box in 0 of 547 held-out images, and 0 of 150 across both fine-tuning arms.
- Predicted the intervention's per-class effect from pre-intervention slopes: it held for the appearance-defined class (Consolidation, predicted +0.018 vs observed +0.015, residual $p = 0.57$) and failed for the geometry-defined one (Cardiomegaly, residual -0.050 , $p < 0.0001$).
- Benchmarked against MC-Dropout logit variance and Grad-CAM under dropout using Friedman tests, BH-corrected Wilcoxon and rank-biserial effect sizes; after correcting a supervision confound in an earlier comparison, MC-Dropout still won in 4 of 5 classes (Cardiomegaly $r = -1.00$). The residual gap tracks label supervision, not image exposure.
- Ran a pre-registered 5-seed replication that retracted one of the earlier significant findings: training variability (SD 0.038) exceeded test-sampling variability (SD 0.028) and all 5 classes flipped sign across seeds.
- Engineered the pipeline for reproducibility on session-limited GPUs without persistent storage: pinned weights for byte-identical initialisation, contamination assertions that halt the run, and incremental artifact publishing so a timeout cost one run instead of ten.

AI Developer Intern — Armour Group (Clinical Form Extraction Pipeline) | May 2026 – Sep 2026

Stack: Python, microsoft/mai-transcribe-2 (speech-to-text), openai/gpt-6-astra-pro with Claude Opus 5 fallback | Built on the Patients Over Paperwork prototype (Armour Group Challenge, June 2026)

- Took a hackathon prototype through to implementation: the extraction pipeline was merged into the product codebase and submitted for a clinical pilot, built cross-functionally with clinicians and engineers, who set the review thresholds and failure cases.

- Extended a six-stage multimodal pipeline that turns multilingual consultation audio into structured clinical forms (11+ form types, 38+ languages), adding a second-pass reviewer for fields lost at segmentation boundaries and an evidence-verification stage for every extracted value.
- Rebuilt the confidence engine around asymmetric clinical cost, cutting Brier score from 0.276 to 0.100 on the reference recording and raising high-confidence-tier precision to 0.93 across two recordings and 0.86 across five.
- Measured the veto layer directly: it removes 60% of extracted fields and 72% of the errors, a 67% hit rate against a 48% base error rate.
- Implemented evidence grounding: each field cites a verbatim transcript span scored for presence, entailment and uniqueness, so ambiguous spans such as a bare “yes” are discounted.
- Migrated speech-to-text to a dedicated transcription endpoint and split the model capability flag into separate temperature, structured-output and token-limit controls, fixing reasoning-model calls that had been dropping JSON mode silently.
- Built the batch evaluation harness and a threshold-fitting tool sweeping false-auto-fill rate against coverage; fixing a floating-point bug there exposed the cleanest operating point, where unanimous signal agreement gave 18 auto-fills with 1 error.
- Added transcription caching with deterministic sampling, so evaluation runs repeat exactly and unchanged stages cost no further inference.

Physics MS Thesis & Post-MS Research Intern — Particle Identification in the CMS HGCAL using Machine Learning | May 2021 – June 2024

Indian Institute of Science Education and Research(IISER) Kolkata & Saha Institute of Nuclear Physics | Report:

https://drive.google.com/file/d/18htvrySMMo9xmfjpaMD3xH_LoDqZASFES/view

Stack: PyTorch, PyTorch Geometric, scikit-learn, SciPy, NumPy, pandas, netgraph, Geant4 | Data: ~100,000 simulated events, 8 classes | Compute: multi-GPU Tesla P100

- Trained Artificial Neural Networks and Graph Neural Networks classifiers on roughly 100,000 simulated detector events across 8 classes for particle identification from shower shapes in the CERN Compact Muon Solenoid HGCAL, reaching 94.76% (ANN) and 96.66% (GNN) accuracy on a multi-GPU Tesla P100 setup.
- Built the standalone Geant4 detector simulation that generated the training data, and the graph construction over calorimeter hits that fed the GNN models.
- Implemented several GNN variants across a two-year post-MS appointment, including residual and attention architectures; residual connections raised accuracy in both the ANN and the GNN, and added detector-noise simulation kept accuracy up under noisier inputs.
- Presented the work as a poster, “Particle Identification based on Shower Shapes in CMS using Machine Learning Algorithms”, at the IISER Kolkata Department of Physical Sciences foundation day (March 2023).

Generative Engine Optimisation Analyst — GradCore (client: Radiology Seminars) | June 2026

Stack: Python, Google Search Console, ChatGPT, Claude Sonnet 5, Gemini Pro, Microsoft 365 Copilot

- Benchmarked LLM retrieval by running matched prompts through four assistants and comparing the provider sets returned against Google’s results. Every model named globally known institutions and dropped local providers holding page-one rank, so search rank did not predict presence in generated answers.
- Built a Python query-mining and gap-classification pipeline over 120 queries from 13 seed topics, labelling each against a four-competitor benchmark by fixed rules (71 flagged, 59.2%) and logging every prompt, model output and source for audit.
- Compiled and delivered the engagement’s 83-page report and 27-slide presentation, working directly with the client throughout.

PROJECTS

Jaffle Demand Lab — Forecasting Pipeline with Ground-Truth Model Evaluation | September 2026

Stack: Python, dbt (Fusion), DuckDB, BigQuery, statsmodels, scikit-learn, LightGBM, pytest, GitHub Actions | Code:

https://github.com/pranavqc/UOB_dbt_workshop_starter

- Built a reproducible forecasting system end to end: a seeded simulator (687k orders, 984k order lines, 1,096 days), a 23-model dbt layer carrying 114 data tests, and a Python model suite that runs from a clean clone on DuckDB without cloud credentials.
- Cut RMSE 38.2% below baseline on a 90-day holdout scored once, benchmarking 9 models over 52,310 store × SKU × day rows under rolling-origin cross-validation, and replacing a Poisson fit with a Negative Binomial GLM after measuring overdispersion (Pearson $\chi^2/df = 4.80$).
- Measured validation leakage instead of asserting it: random K-fold overstated accuracy by 6.1% against a size-matched rolling fold, with point-in-time correctness enforced by a dbt test that recomputes every rolling window. Five injected source faults each pair a detecting test with a repairing model, under CI on every commit.

Predictive Modelling for UK Restaurant Insolvency | Jan – March 2026

Stack: Python, XGBoost (CUDA A100), scikit-learn, imbalanced-learn (SMOTE), Hugging Face Transformers, Google Places, pandas, Folium |

Code+Report: https://github.com/pranavqc/Spill_the_tea

- Built an end-to-end pipeline processing 5.6M+ Companies House records into a 24,750-sample dataset, resolving entity matching across four sources with no shared primary key.
- Applied zero-shot sentiment classification (facebook/bart-large-mnli) and trained an XGBoost classifier (1,200 trees, CUDA A100) reaching F1 = 0.24, precision = 0.48 and AUC = 0.73 on the minority class under 95:5 imbalance.
- Showed via gain-based feature importance that structural and administrative signals beat customer-facing metrics as failure predictors, and are available before a restaurant opens.

SkillSync — AI-Powered Career Intelligence Platform | April – May 2026

Stack: Python, LangChain, Gemini Pro, Mistral-7B, FastAPI, Firestore, BigQuery, Docker, React/TypeScript | Code:

<https://github.com/pranavqc/SkillSync>

- Built a multi-agent LangChain pipeline (skill extraction, then live job-market research and gap analysis) with Pydantic-constrained LLM decoding producing deduplicated skill taxonomies, deployed via async FastAPI, BigQuery storage and multi-stage Docker builds on Cloud Run.

Other Projects

- Muon Scattering Tomography (2023): U-Net CNN segmentation to localise civil structures and materials from Geant4 muon-scattering simulations.
- MLOps & Deployment (2024): automated CI/CD ML pipelines (GitHub Actions, Hugging Face) and a deployed Gradio chatbot.

CERTIFICATIONS

Generative AI with Diffusion Models (NVIDIA, 2025) | GANs, CNNs and Sequence Models (DeepLearning.AI, 2023–2024) | Machine Learning Specialization (Coursera, 2023) | LangChain Agentic AI and LangSmith Workshop (PwC, 2026)

ADDITIONAL

Teaching Assistant, first-year physics laboratory — IISER Kolkata (Dec 2021 – Apr 2022). Volunteering: UoB Xtra Campus 3K and Brum Biosphere (2026).